Chapter 8

METALLURGY

1. Write names.

a. Alloy of sodium with mercury.

Sodium amalgam

b. Molecular formula of the common ore of aluminium.

$Al_2O_3.nH_2O$

c. The oxide that forms salt and water by reacting with both acid and base.

Aluminium oxide, Zinc oxide (Amphoteric oxides)

d. The device used for grinding an ore.

Ball mill

e. The nonmetal having electrical conductivity.

Graphite (allotrope of carbon)

f. The reagent that dissolves noble metals.

Aqua regia

2. Make pairs of substances and their properties

Substance	Property
Potassium bromide	Soluble in water
Gold	High ductility
Sulphur	Combustible
Neon	No chemical reaction

3. Identify the pairs of metals and their ores from the following.

Group A	Group B
Bauxite	Aluminium
Cassiterite	Tin
Cinnabar	Mercury

4. Explain the terms.

a. Metallurgy

The science and technology regarding the extraction of metals from ores and their purification for the use is called metallurgy.

b. Ores

The minerals from which the metals can be separated economically are called ores.

c. Minerals

The compounds of metals that occur in nature along with the impurities are called minerals.

d. Gangue

Ores contain metal compounds with some of the impurities like soil, sand, rocky substances etc. These impurities are called gangue.

5. Write scientific reasons.

a. Lemon or tamarind is used for cleaning copper vessels turned greenish.

i. Copper on exposure to moist air combines with carbon dioxide and a green-coloured copper carbonate salt is formed, tarnishing the copper.

ii. Lime juice and tamarind contain citric acid and tartaric acid respectively.

iii. These acids react with copper carbonate forming water - soluble salts, which get washed off easily.

iv. Due to this, the tarnished copper becomes clean and regains its lustre.

v. Hence, lemon or tamarind is used for cleaning copper vessels turned greenish.

b. Generally, the ionic compounds have high melting points.

i. Ionic compounds form when an atom of a metallic element in combination transfers its electrons to an atom of another non-metallic element.

ii. In this process, the metallic atom, after losing electrons, becomes a positively charged ion known as a cation, while the non-metallic atom, after gaining electrons, becomes a negatively charged ion known as an anion.

iii. These ions are held together by strong electrostatic forces of attraction.

iv. Considerable amount of energy is required to break these strong forces of attraction. Hence, the melting points of ionic compounds are high.

c. Sodium is always kept in kerosene.

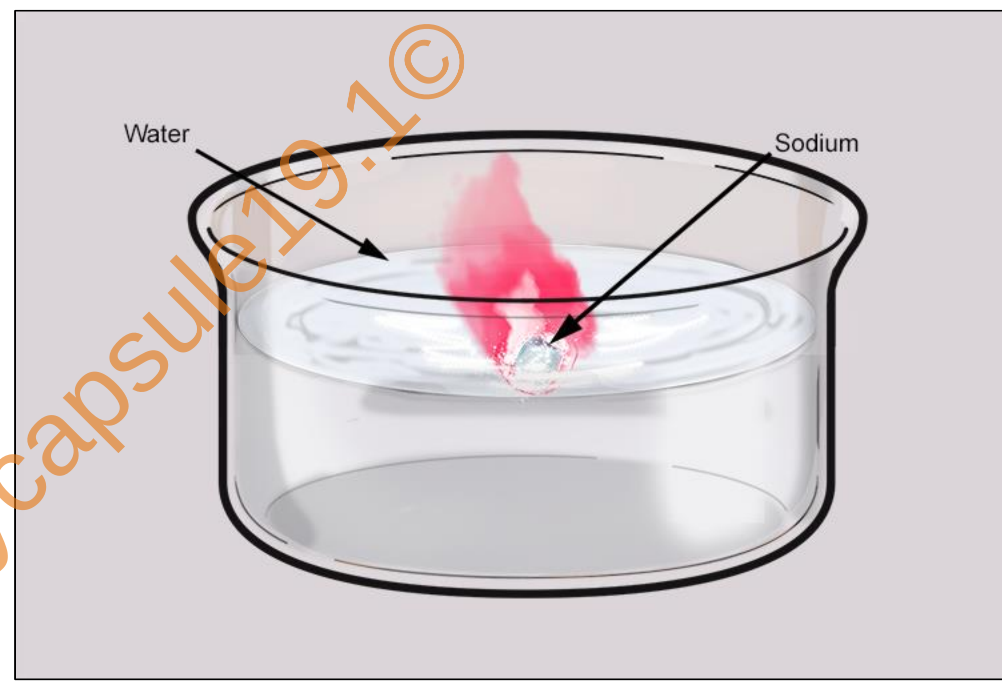
i. Sodium is a highly reactive metal.

ii. It reacts with the oxygen and moisture present in the air to form sodium hydroxide and hydrogen gas.

iii. Hydrogen which is released, catches fire in presence of oxygen due to liberation of heat during this reaction.

iv. Sodium does not react with kerosene and hence, it sinks in it.

v. Thus, to prevent sodium from coming in contact with air, it is immersed in kerosene.



d. Pine oil is used in froth flotation.

i. Pine oil is hydrophobic.

ii. The particles of the metal sulfide ores that are hydrophobic become wetted by pine oil.

iii. They float with the foam on the surface of the water.

iv. Thus, by utilizing this property, some ores containing hydrophobic particles can be concentrated through the froth flotation process. Therefore, pine oil is used in froth flotation.

e. Anodes need to be replaced from time to time during the electrolysis of alumina.

i. The cathode and anodes are made of graphite.

ii. During the electrolysis process, aluminium is deposited on the cathode.

iii. Oxygen is liberated at the anode.

iv. Some of this oxygen reacts with the carbon in the graphite to form carbon - dioxide gas.

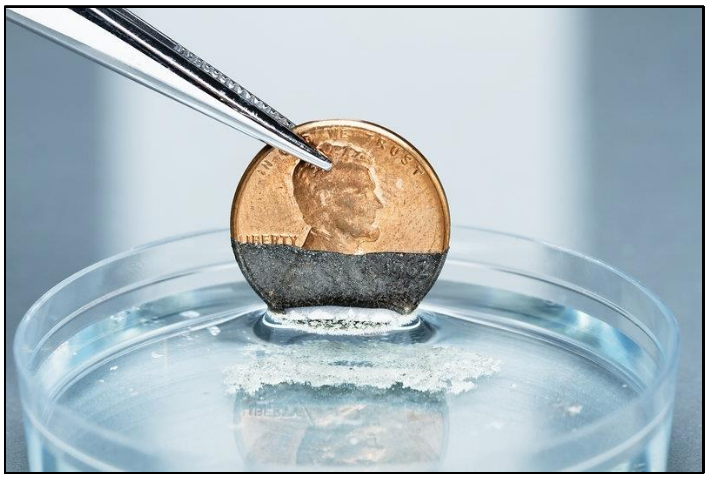
v. Thus, the anodes need to be replaced from time to time as they get oxidised during the electrolysis of alumina.

6. When a copper coin is dipped in silver nitrate solution, a glitter appears on the coin after some time. Why does this happen? Write the chemical equation.

Copper being more reactive than silver, displaces silver from silver nitrate to form element silver and copper nitrate.

$$\underset{\text{Copper}}{Cu_{(s)}} + \underset{\text{Silver nitrate}}{2AgNO_{3(aq)}} \rightarrow \underset{\text{Copper nitrate}}{Cu(NO_3)_{2(aq)}} + \underset{\text{Silver}}{2Ag_{(s)}}$$

The displaced silver gets coated on to the coin, hence coin gets the silver shine.



7. The electronic configuration of metal 'A' is 2,8,1 and that of metal 'B' is 2,8,2. Which of the two metals is more reactive? Write their reaction with dilute hydrochloric acid.

Metal 'A' Electronic configuration - (2,8,1),

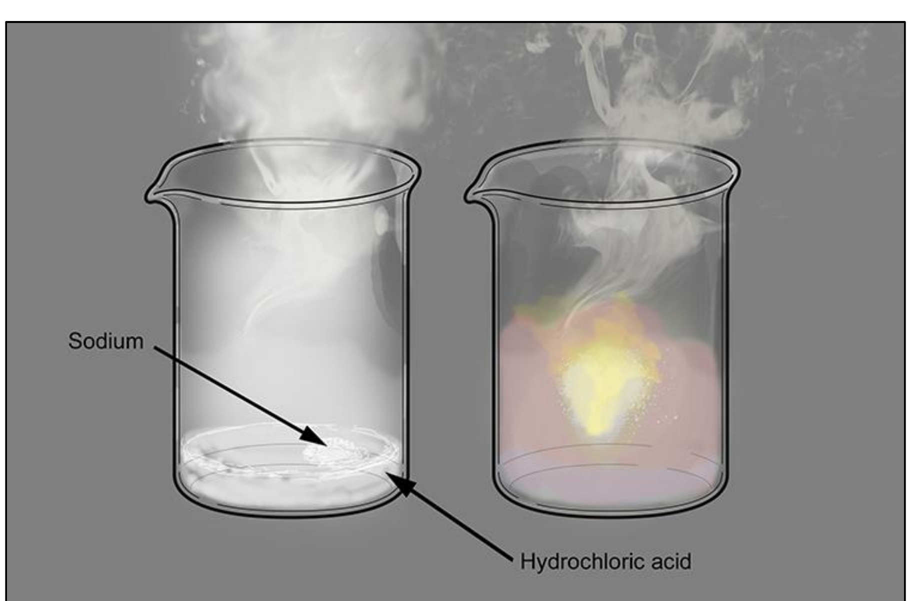
Metal 'B' Electronic configuration - (2,8,2),

Metal 'A' loses its one electron and form positive ion. Metal 'B' also loses 2 electrons and forms positive ion; but metal 'A' is more reactive than metal 'B', as metal 'B' takes more energy to remove two electrons where as metal 'A' requires less energy to lose one electron.

Metal 'A' is sodium and metal 'B' is magnesium. Hence, sodium is more reactive than magnesium.

$$\underset{\textit{Sodium}}{2\,Na} + \underset{\textit{Hydrochloric acid}}{2HCl} \rightarrow \underset{\textit{Sodium chloride}}{2NaCl} + \underset{\textit{Hydrogen gas}}{H_2\uparrow}$$

$$\underset{\textit{Magnesium}}{Mg} + \underset{\textit{Hydrochloric acid}}{2HCl} \rightarrow \underset{\textit{Magnesium Chloride}}{MgCl_2} + \underset{\textit{Hydrogen gas}}{H_2\uparrow}$$



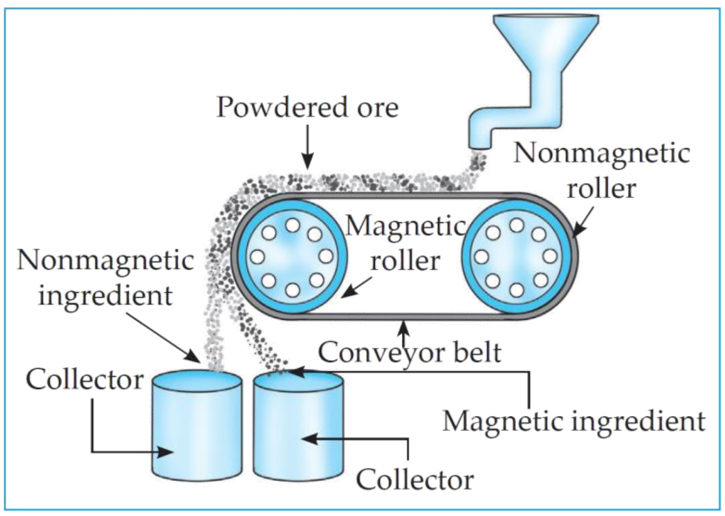
8. Draw a neat labelled diagram.

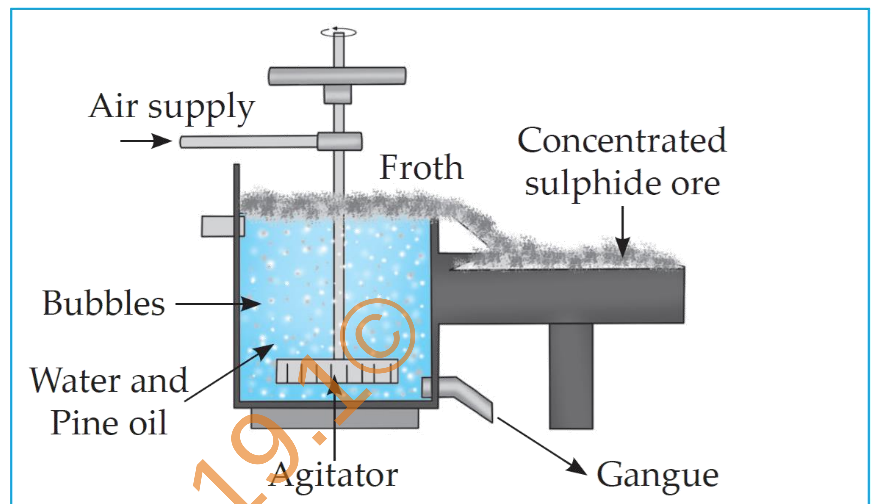
a. Magnetic separation method.

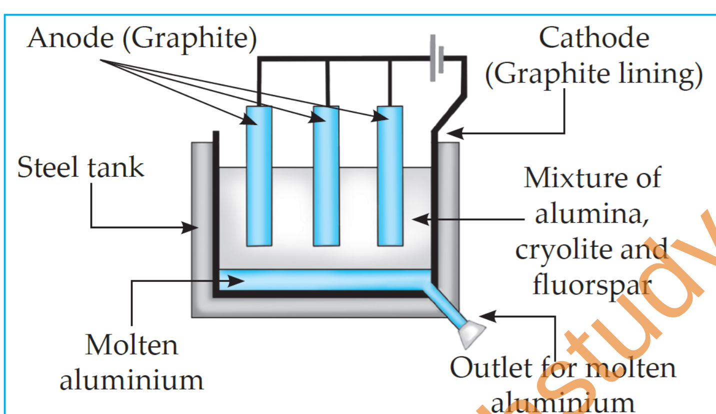
b. Froth floatation method.

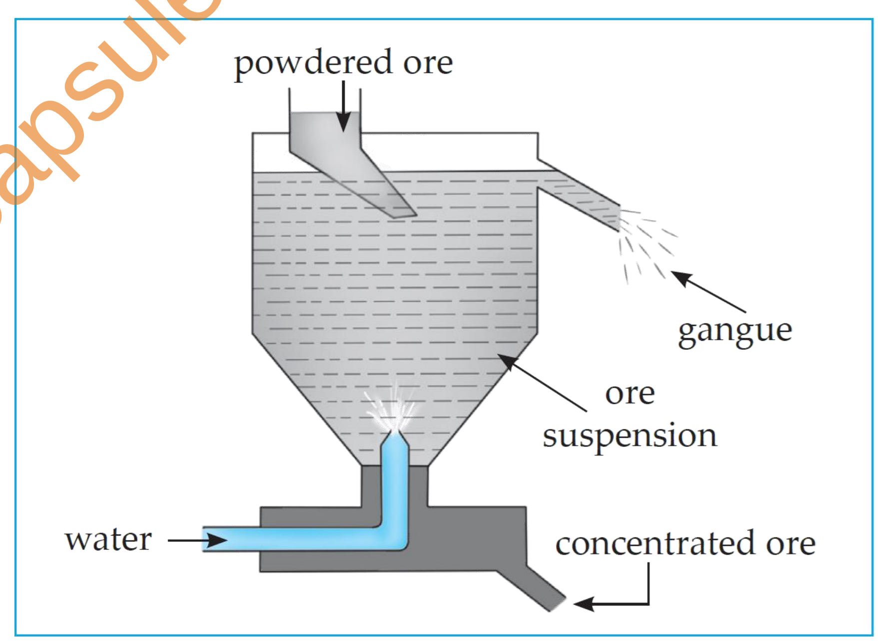
c. Electrolytic reduction of alumina.

d. Hydraulic separation method.









9. Write chemical equation for the following events.

a. Aluminium came in contact with air.

Aluminium combines with oxygen to form aluminium oxide.

$$4Al + 3O_2 \rightarrow 2\,Al_2O_3$$

Aluminum	*Oxygen*		*Aluminium Oxide*

b. Iron filings are dropped in aqueous solution of copper sulphate.

When iron filings are dropped in aqueous solution of copper sulphate, iron displaces copper from the solution. The blue colour of copper sulphate solution fades out.

Fe	+	$CuSO_4$	$\rightarrow$	$FeSO_4$	+	Cu
Iron		*Copper Sulphate*		*Iron Sulphate*		*Copper*

c. A reaction was brought about between ferric oxide and aluminium.

When Ferric oxide reacts with aluminium, aluminium oxide and iron are formed. This reaction is known as "Thermit Reaction".

Fe_2O_3	+	$2Al$	$\rightarrow$	$2Fe$	+	$Al_2O_3 + \text{heat}$
Ferric Oxide		*Aluminium*		*Iron*		*Aluminium Oxide*

d. Electrolysis of alumina is done.

Electrolysis of molten mixture of alumina is done in a steel tank.

Electrode reaction :

Anode reaction $2O^{2-} \rightarrow O_2 + 4e^-$ (Oxidation)

Cathode reaction $Al^{3+} + 3e^- \rightarrow Al$ (Reduction)

e. Zinc oxide is dissolved in dilute hydrochloric acid.

When zinc oxide dissolves in dilute hydrochloric acid, zinc chloride and water is formed.

ZnO	+	$2HCl$	$\rightarrow$	$ZnCl_2$	+	H_2O
Zinc Oxide		*Hydrochloric Acid*		*Zinc Chloride*		*Water*

10. Complete the following statement using every given options.

During the extraction of aluminium………….

a. Ingredients and gangue in bauxite.

Aluminium is extracted from its ore Bauxite ($Al_2O_3.nH_2O$).

Bauxite contains 30% to 70% of Al_2O_3 and remaining part is gangue. It is made up of sand, silica, iron oxide etc. Silica (SiO_2), ferric oxide (Fe_2O_3) and titanium oxide (TiO_2) are the impurities present in bauxite.

b. Use of leaching during the concentration of ore.

i. Leaching is a method in which ore is soaked in a certain solution for a long time.

ii. The ore dissolves in that solution due to a specific chemical reaction.

iii. The gangue does not react and therefore does not dissolve in that solution.

iv. During the concentration of ore, separation of impurities is done by leaching process using either Bayer's method or Hall's method.

c. Chemical reaction of transformation of bauxite into alumina by Hall's process.

i. In the Hall's process, the ore is powdered and then leached by heating with aqueous sodium carbonate to form water soluble sodium aluminate.

ii. Then the insoluble impurities are filtered out.

iii. The filtrate is warmed and neutralised by passing carbon dioxide gas through it.

iv. This results in the precipitation of aluminium hydroxide.

v. $Al_2O_3.2H_2O_{(s)} + Na_2CO_{3(aq)} \longrightarrow 2NaAlO_{2\ (aq)} + CO_{2\ (g)} + 2H_2O_{(l)}$

vi. $2NaAlO_{2(aq)} + 3H_2O_{(l)} + CO_{2(g)} \longrightarrow 2Al(OH)_3\downarrow + Na_2CO_3$

vii. The precipitate of $Al(OH)_3$ obtained in Hall's process is filtered, washed, dried and then calcinated by heating at 1000°C to obtain alumina.

viii. $2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O.$

d. Heating the aluminium ore with concentrated caustic soda.

Aluminium oxide being amphoteric in nature, it reacts with the concentrated solution of sodium hydroxide (caustic soda) at 140°C to 150°C under high pressure for 2 to 8 hours in a digester to form water soluble sodium aluminate.

$$\underset{\text{Aluminium oxide}}{Al_2O_3.2H_2O_{(s)}} + \underset{\text{Sodium hydroxide}}{2NaOH_{(aq)}} \rightarrow \underset{\text{Sodium aluminate}}{2NaAlO_{2\ (aq)}} + \underset{\text{Water}}{3H_2O_{(l)}}$$

11. Divide the metals Cu, Zn, Ca, Mg, Fe, Na, Li into three groups, namely reactive metals, moderately reactive metals and less reactive metals.

Reactive metals	Moderately reactive	Less reactive
Na	Mg	Cu
Li	Zn	
Ca	Fe	

INTEXT QUESTIONS

1. **Which method do we use when we want to study many things together and at the same time?**

   We use classification method if we want to study many things together at the same time.

2. **What are the physical properties of metals and nonmetals?**

   PHYSICAL PROPERTIES OF METALS:

i. Metals are in solid state except Mercury and Gallium which are in liquid state.

ii. Metals are hard, have lustre and high melting and boiling points. They are ductile, malleable, good conductors of heat and electricity.

   PHYSICAL PROPERTIES OF NON - METALS:

i. Some non - metals are in solid state while some are in gaseous state.

ii. They do not possess lustre. Non - metals are not hard, except diamond.

iii. They have low melting and boiling points.

iv. Not malleable and not ductile.

v. Non - metals are bad conductors of heat and electricity, except graphite.

3. **In the reaction between chlorine and HBr a transformation of HBr into Br_2 takes place. Can this transformation be called oxidation? Which is the oxidant that brings about this oxidation?**

$$Cl_2\,(g) + 2HBr\,(aq) \longrightarrow 2HCl(aq) + Br_2(aq)$$

The conversion of HBr to Br_2 is an oxidation process. In the above reaction, Cl_2 in the oxidant.

4. What is the electronic definition of oxidation and reduction?

Definition in terms of electron transfer:

i. Oxidation means loss of electrons.

ii. Reduction means gain of electrons.

5. Write the electrode reaction for electrolysis of molten magnesium chloride and calcium chloride.

For Magnesium chloride

$Mg^{2+} + 2e^- \longrightarrow$ Mg(magnesium metal at the cathode)

$2Cl^- - 2e^- \longrightarrow Cl_2$ (chlorine gas at the anode).

Magnesium ions gain electrons (reduction) to form magnesium atoms. Chloride ions lose electrons (oxidation) to form chlorine atoms.

The overall reaction is:

$MgCl_{2\,(l)} \longrightarrow Mg_{(s)} + Cl_{2\,(g)}\uparrow$

For Calcium chloride

$Ca^{2+} + 2e^- \longrightarrow$ Ca (calcium metal at the cathode)

$2Cl^- - 2e^- \longrightarrow Cl_2$ (chlorine gas at the anode)

Calcium ions gain electrons (reduction) to form calcium atoms. Chloride ions lose electrons (oxidation) to form chlorine atoms.

Overall reaction is: $CaCl_2 \longrightarrow Ca_{(s)} + Cl_2\uparrow$

6. What are the moderately reactive metals?

The metals present in the middle of reactivity series are called as, moderately reactive metals e.g., Zinc, Iron, Lead, Tin, Copper.

7. In which form do the moderately reactive metals occur in nature?

These metals are usually present as Sulphides or Carbonates in nature.

8. Collect the information regarding how mercury is extracted from its ore cinnabar and write the corresponding chemical reaction.

Cinnabar is an ore of mercury. When cinnabar is heated (roasted), it is converted into mercuric oxide (HgO). Mercuric oxide is then reduced to mercury on further heating.

$$\underset{\text{Cinnabar}}{2HgS_{(s)}} + 3O_{2(g)} \xrightarrow{\text{heat}} 2HgO_{(s)} + 2SO_{2(g)}$$

$$\underset{\text{Mercuric oxide}}{2HgO_{(s)}} \xrightarrow{\text{heat}} \underset{\text{Mercury}}{2Hg} + \underset{\text{Oxygen}}{O_{2(g)}}$$

9. What is meant by corrosion?

The degradation of materials due to reaction with the environment is called corrosion.

10. Why do silver articles turn blackish while copper vessels turn greenish on keeping in air for a long time?

i. On exposure to air, Silver articles turn blackish after some time because of the layer of silver sulphide (Ag2S) formed by the reaction of silver with hydrogen sulphide in the air.

ii. Carbon dioxide in moist air reacts with the surface of copper vessel and these copper vessels turn greenish because of the formation of copper carbonate ($CuCO_3$) on its surface.

11. Why do pure gold and platinum always glitter?

i. Gold and Platinum occur in free state.

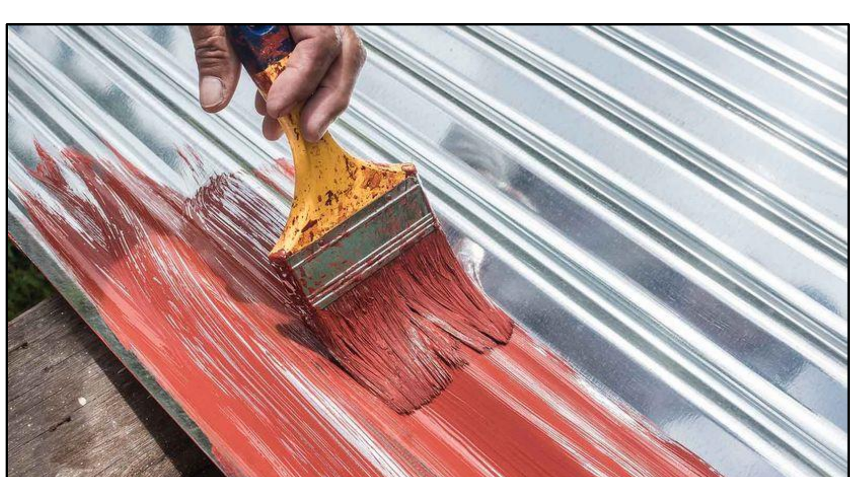
ii. They are noble metals; hence they are least reactive.

iii. They are not affected by air, water and other natural factors.

iv. Hence, pure gold and platinum always glitter.

12. Which measures would you suggest to stop the corrosion of metallic articles or not to allow the corrosion to start?

You can prevent the corrosion of the metal by coating their surface using any of the following: By applying oil, grease, paint or varnish on the surface. By coating/depositing a thin layer of any other metal which does not corrode.



13. What is done so to prevent rusting of iron windows and iron doors of your house?

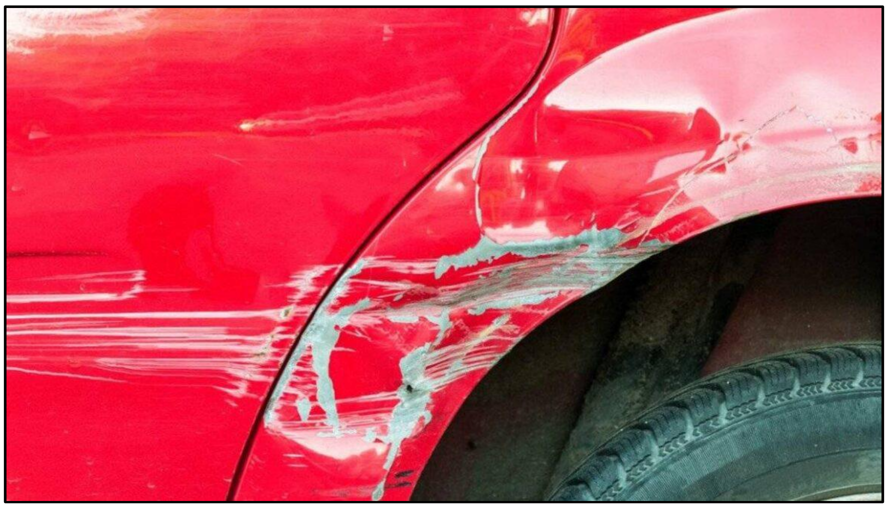
We can lower the rate of the process of rusting of iron by applying a layer of paint, oil, grease or varnish on their surface. Thus, we can prevent rusting of iron windows and doors of our house.

14. Can we permanently prevent the rusting of an iron article by applying a layer of paint on its surface?

i. No, we cannot protect the articles permanently from rusting by painting on them.

ii. The method of painting is suitable for short time.

iii. If there is a scratch on the paint on the surface of the article and if a small surface of the metal comes in contact with air, the process of rusting starts below the layer of the paint.



15. What are the various alloys used in daily life? Where are those used?

i. Brass and Bronze are the two alloys which we use in our daily life.

ii. Brass is an alloy of copper and zinc, and Bronze is an alloy of copper and tin.

iii. Brass is used for decorative articles and for making musical instruments.

iv. Bronze is used for making medals, sculptures, statues cooking utensils and coins.

16. What are the properties that the alloy used for minting coins should have?

Alloys used for minting coins should be ductile and malleable.

Previous Year Questions

1. Out of sodium and sulfur which is a metal? Explain its reaction with the oxygen.

Out of sodium and sulphur, sodium is a metal. So, sodium reacts with oxygen to form sodium oxide.

$$4\,Na_{(s)} + O_{2\,(g)} \rightarrow 2\,Na_2O_{(S)}$$

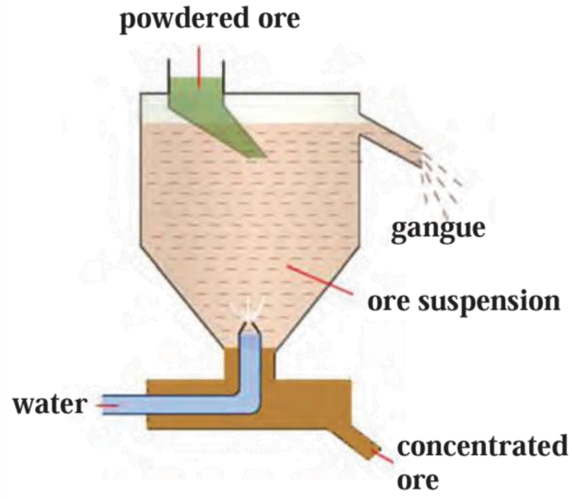
Sodium Oxygen Sodium oxide

2. A tapping vessel opens in a tank like container that is tapering on the lower side. The tank has an outlet for water on the upper side and a water inlet on the lower side. Finely ground ore is released in the tank. A forceful jet of water is introduced in the tank from lower side and gangue particles and pure ore are separated by this method.

i. The above description is of which gravitation separation method?

The above description is of Hydraulic separation method.

ii. Draw labelled diagram of this method.



3. In two methods of control of corrosion of aluminium, either a layer of aluminium oxide is formed, or a silver plating is done on the surface. State to which electrode the aluminium article is attached in these methods, respectively.

i. In the method of forming a layer of aluminum oxide to control corrosion of aluminum, the aluminum article is attached to the anode (positive electrode) of an electrolytic cell.

ii. In the method of silver plating to control corrosion of aluminum, the aluminum article is attached to the cathode (negative electrode) of an electrolytic cell.

4. Anodes need to be replaced from time to time during the electrolysis of alumina. – Exercise Question 5 - e

5. Write a short note on Alloying.

i. Majority of the metallic substances used presently are in the form of alloys.

ii. The main intention behind this is to decrease the intensity of corrosion of metals.

iii. The homogenous mixture formed by mixing a metal with other metals or nonmetals in certain proportion is called an alloy.

iv. For example, bronze is an alloy formed from 90% copper and 10 % tin. Bronze statues do not get affected by sun and rain.

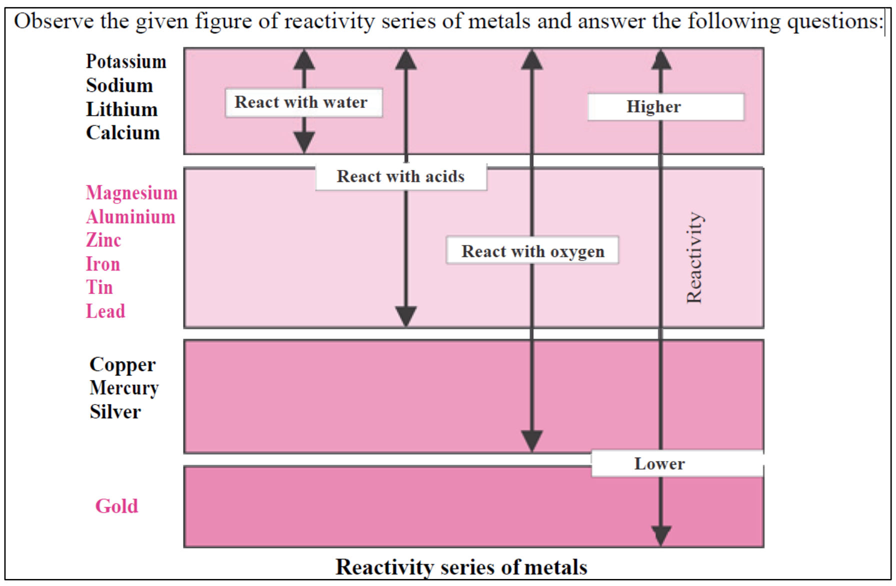
v. Stainless steel does not get stains with air or water and does not rust. It is an alloy made from 74% iron, 18% chromium and 8% carbon.

5. Observe the given figure of reactivity series of metals and answer the following questions:

a. Name the two metals which react with water.

b. Name the two moderately reactive metals.

c. Name the most highly reactive metal and the most less reactive metal.



a. K, Na

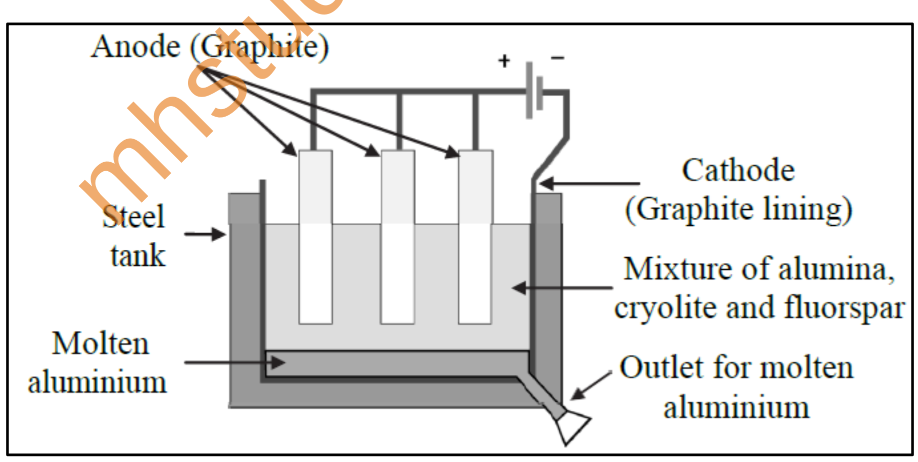
b. Zn, Fe

c. Highly reactive – K,

Least reactive – Au

6. The copper vessels turn greenish and silver articles turn blackish when kept open in air for long time. – Intext Question

7. Observe the following diagram and answer the questions.



a. Write the 'anode reaction'.

b. Write the 'cathode reaction'.

c. What is the purpose of mixing 'cryolite' and 'fluorspar' with 'alumina' in the electrolytic reduction of alumina?

a. Anode reaction:

$2O^{2-} \rightarrow O_2 + 4e^-$ (Oxidation)

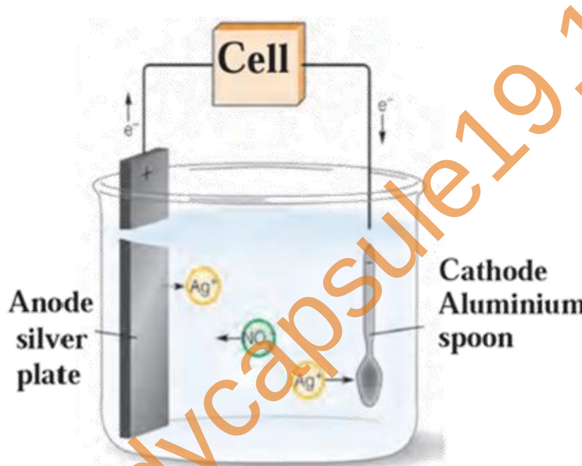
b. Cathode reaction:

$Al^{3+} + 3e^- \rightarrow Al_{(l)}$ (Reduction)

c. Cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added in the mixture of alumina to lower its melting point up to 1000 ^{0}C.

8. Sodium metal is always kept in kerosene. – Exercise Question 5 - c

9. Identify the process described in the sketch and write any two uses.



i. The process described in the sketch is electroplating

ii. In this method a less reactive metal is coated on a more reactive metal by electrolysis. Silver plated spoons, gold plated ornaments are the examples of electroplating.

10. The electronic configuration of metal 'A' is 2,8,1 and that of metal 'B' is 2,8,2. Which of the two metals is more reactive? Write their reaction with dilute hydrochloric acid. – Exercise Question 7

11. (a) Name the main ore of aluminium. (b) What impurities are present in aluminium ore?

a. Bauxite is the main ore of aluminium.

b. Silica (SiO_2), ferric oxide (Fe_2O_3) and titanium oxide (TiO_2) are the impurities present in bauxite.

12. Explain the following terms: (a) Metallurgy (b) Ores (c) Gangue. – Exercise Question 4

13. Lemon or tamarind is used for cleaning copper vessels turned greenish. – Exercise Question 5 - a

14. Read the following sentence and answer the questions: "NaCl is an ionic compound." (a) Why is NaCl an ionic compound? (b) State any two properties of ionic compounds.

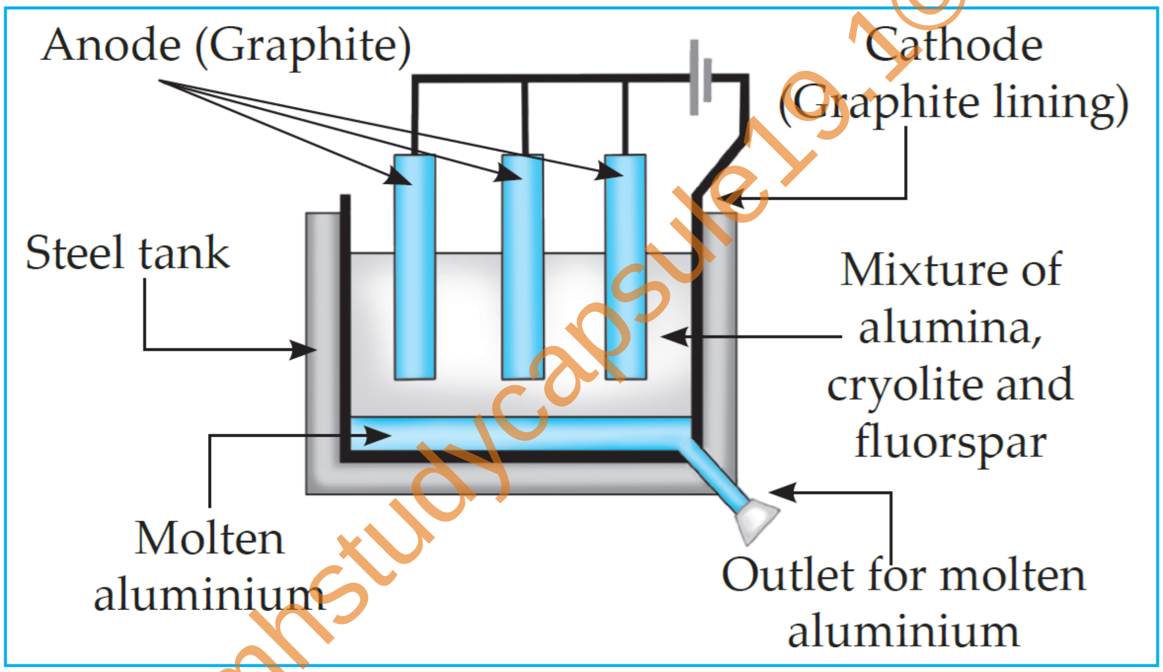
a. Chlorine has an electronic configuration of (2, 8, 7) while sodium has a configuration of (2, 8, 1). Sodium atom donates one of its electrons to form positively charge ion called cation. Chlorine accepts the electron and it and forms negatively charged ion called anion. These ions are held together by strong electrostatic forces of attraction. Thus, it is an ionic compound because sodium chloride is created by the exchange of electrons between the atoms.

b. Two properties of ionic compounds:

1. Ionic compounds have high melting as well as boiling points.
2. They are Good Insulators, hard and brittle in nature.

15. Identify the process given below and accordingly draw neat labelled diagram: A molten mixture of alumina (melting point > 2000°C) is done in a steel tank. The tank has a graphite lining on the inner side. The lining does the work of cathode. A set of graphite rods dipped in the molten electrolyte works as anode. Cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added in the mixture to lower its melting point upto 1000°C.

The process in the paragraph is electrolytic reduction of alumina.



16. Explain the term anodization with example. Give one use of it.

a. In this method metals like copper, aluminium are coated with a thin and strong layer of their oxides by means of electrolysis. For this the copper or aluminium article is used as anode. As this oxide layer is strong and uniform all over the surface, it is useful for prevention of the corrosion of the metal.

For example, when aluminium is anodised, the thin layer of aluminium oxide is formed. It obstructs the contact of the aluminium with oxygen and water. This prevents further oxidation.

b. Use – Manufacturing of anodised pressure cooker.